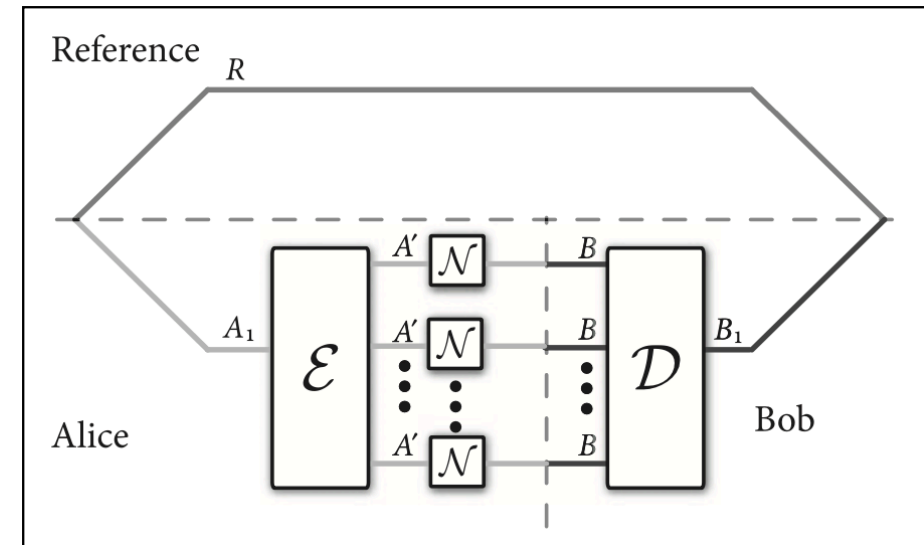




Quantum Capacity

- Alice would like to transmit quantum information intact to or generate entanglement with Bob, perhaps for some distributed quantum computation.
- Alice begins with some shared entanglement with a reference system R in the form of some pure state φ_{RA} .
- She wishes to transfer the correlation between her system and R to Bob, to produce a state ω_{RB} which is close to φ_{RA}
- Quantum Capacity (qubits/channel use) : Maximum of Coherent Information:

- $Q(\mathcal{N}) \geq \max_{\{\varphi_{AA'} \text{ pure} \}} I(A \rangle B)_\rho$
- $I(A \rangle B)_\sigma = H(B)_\sigma - H(AB)_\sigma = H(B)_\sigma - H(E)_\sigma.$



Quantum Capacity of Identity Channel



12-Jul-2026

Quantum Capacity of Identity Channel.

Main Channel: $N(p) = p$

Complementary Channel: $N'(p) = 1 \otimes 0$

$$\begin{aligned} Q(N) &= H(N(p)) - H(N'(p)) \\ &= H(p) - \underbrace{H(1 \otimes 0)}_{=0} \end{aligned}$$

$$Q(N) = \max H(p)$$

$$\Rightarrow H(p) \leq \log_2 2 = 1.$$

$$p = 1/2$$

Quantum Cap. = 1 Qubit/ch. use.

For Noiseless Channel: $C = 1 \text{ bit/ch. use.} \Rightarrow C_E = 2.$

$$P = 1 \text{ bit/ch. use}$$

$$Q = 1 \text{ Qubit/ch. use}$$

References

1. R. Wolf, *Quantum Key Distribution: An Introduction with Exercises*, Lecture Notes in Physics, vol. 988. Switzerland: Springer, 2021.
2. G. Cariolaro, *Quantum Communications*. Switzerland: Springer, 2015.
3. M. M. Wilde, *Quantum Information Theory*. Cambridge University Press, 2013.



Thank You



CEPIITD

Online PG Diploma: Advanced Communication Engineering with Quantum and AI Integration

Course: Quantum Information and Security

Instructor: Dr. Neel Kanth Kundu

Assistant Professor

Centre for Applied Research in Electronics (CARE)
Bharti School of Telecom Tech. & Mgmt. (Joint Faculty)
Indian Institute of Technology Delhi



Quantum Information and Security

Module-3

Quantum Key Distribution

Instructor: Dr. Neel Kanth Kundu

Assistant Professor

Centre for Applied Research in Electronics (CARE)

Bharti School of Telecom Tech. & Mgmt.

Indian Institute of Technology Delhi

Overview



- Introduction
- Quantum Key Distribution (QKD)
- Prepare and Measure QKD Protocols
- Entanglement Based QKD Protocols
- CV-QKD Protocol



Classical Cryptography

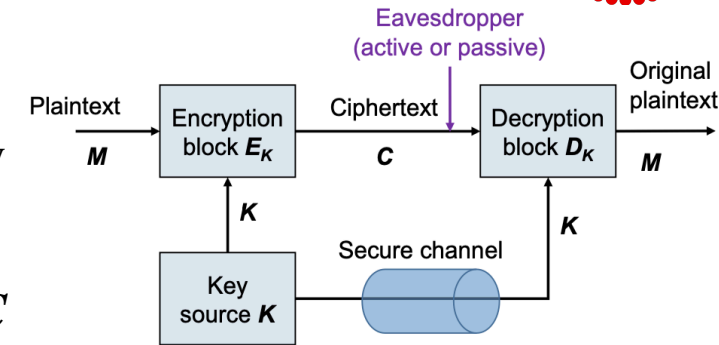
- Cryptography is the art of encoding messages to keep them secret.
- Classical cryptography uses fixed rules (algorithms) to encrypt and decrypt.
- **Goal:** Only sender and receiver can read the message.
- But classical methods have serious limitations in today's world.
- Why Does Classical Cryptography Fail?
 - **Deterministic encryption:** Same input $\rightarrow$ same output.
 - **Key reuse:** Makes it easier to detect patterns.
 - **Frequency analysis:** Attackers can analyze letter frequency to guess plaintext.
 - **Brute-force attacks:** Try every possible key (especially easy with short keys).
 - **Not information-theoretically secure:** Can be broken with enough computing power.
- **Information Theoretic Security:** Impossible to break—even in theory—without the key. Remains secure even against all-powerful adversaries.

Conventional Cryptography

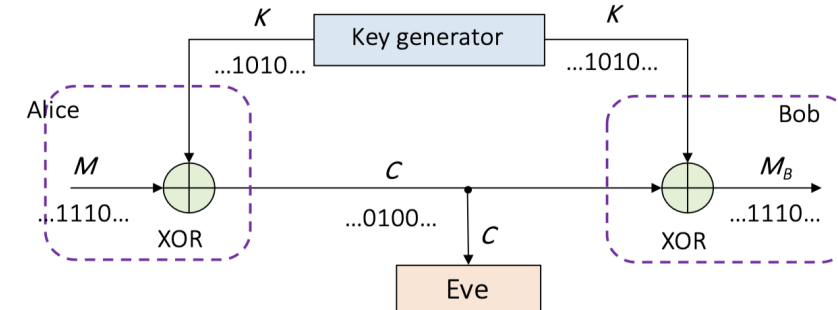


- Key Based Crypto systems:
 1. Symmetric Algorithms: Same key used for encryption & decryption
 2. Asymmetric Algorithms: Public encryption key & private decryption key
- Simplest private key cryptosystem: One time pad
- Unconditional (Information theoretic) Security: Message M and cryptogram C are statistically independent i.e.,

$$I(M, C) = H(M) - H(M|C) = 0 \Leftrightarrow H(M|C) = H(M)$$
- Condition for perfect secrecy: $H(M) \leq H(K)$
- Computational Security:
 1. Security is guaranteed when Eve runs cryptanalytic attack for limited amount of time. Security can be broken if sufficient computational resource and time are available.
 2. Eve can break the security with small probability of error.



Basic key-based cryptographic scheme

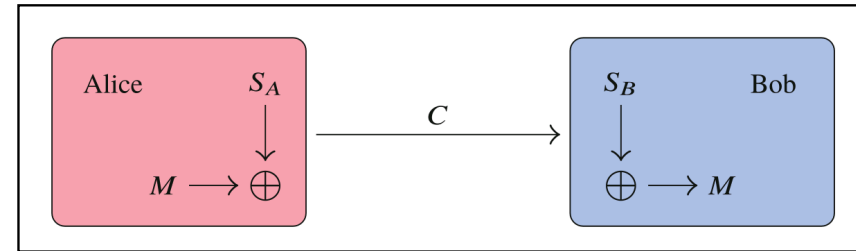


One-time pad encryption scheme



One Time Pad

- Two parties, Alice and Bob, want to communicate securely.
- An adversary Eve can listen to everything but cannot alter messages.
- Alice and Bob share a secret random key $S_A = S_B \in \{0,1\}^n$.
- Message $M = (M_1, \dots, M_n) \in \{0,1\}^n$.



One-time pad encryption.

- **Goal:** Ensure Eve learns nothing about M .
- **Encryption:** Alice encrypts the message M using her key S_A : $C = M \oplus S_A$.
- **Transmission:** The ciphertext C is sent to Bob over a public channel.
- **Decryption:** Bob decrypts the message using his key S_B : $M' = C \oplus S_B$.

$$\begin{aligned} M'_i &= C_i \oplus (S_B)_i \\ &= (M_i \oplus (S_A)_i) \oplus (S_B)_i = M_i \oplus ((S_A)_i \oplus (S_B)_i) \\ &= M_i \oplus 0 \quad (\text{since } S_A = S_B) \\ &= M_i \end{aligned}$$

• **Correctness of Decryption:**



One Time Pad

- **Perfect secrecy** proved by Claude Shannon (1949)
- **Requirements** for this protocol to be information-theoretically secure:
 - Key must be **truly random** (uncorrelated bits)
 - Key length $\geq$ message length
 - Key must **never be reused**

$$C_1 = M \oplus S$$

- If same key S encrypts two messages M, N :

$$C_2 = N \oplus S$$

$$C_1 \oplus C_2 = M \oplus N$$

- Eve learns relationship between messages!
- Key must be **completely secret**
- **Limitations of One-Time Pad:**
 - Requires a truly random key S as long as the message.
 - Key must be **used only once** (hence the name).
 - Secure key distribution and storage is challenging.
 - Not practical for many modern applications.



Key Generation

- Ideal key generator setting.
- An ideal device outputs two identical, random keys $S_A = S_B$ for Alice and Bob.
- Eve cannot interact with it (i.e., nothing can be input or output to Eve's part of the system, which is denoted by the symbol $\perp$).
- Properties of Practical Key Generation:

- **Correctness:** The probability that Alice and Bob's keys differ but the protocol **does not** abort is small:

$$\Pr[S_A \neq S_B] \leq \epsilon .$$

- **Close to perfect key:**
 - Eve does not have any knowledge of the key.
 - The individual key bits are uncorrelated.
- **Detectable attacks:** Protocol aborts if Eve interferes

